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Late Paleozoic remagnetization in limestones of the Craven Basin (northern England) and the rock magnetic fingerprint of remagnetized sedimentary carbonates

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Abstract. Samples of Lower Carboniferous Chatburn and Pendleside limestones collected from 27 sites in the Craven Basin (northern England) carry a syn-Hercynian remagnetization residing in magnetite. The paleomagnetic directions are all of reversed polarity consistent with a Kiaman Superchron remagnetization. The partially (62%) tilt corrected mean direction has the minimum correlation statistic implying a synfolding magnetization. The resulting pole appears to be Late Carboniferous in age, and there is no indication of any magnetization component that predates the remagnetization event. The magnetization appears to be of the same type as the synfolding remagnetizations documented in several Appalachian carbonate units. Jackson (1990) reported that remagnetized Appalachian carbonates have distinctive hysteresis properties, and we have obtained similar hysteresis loops from remagnetized Paleozoic carbonates from the Craven Basin (England), the Great Basin (Nevada), Alaska, and New York State. The hysteresis loops are all "wasp-waisted" and yield highly anomalous hysteresis ratios, compared to data from synthetic or "typical" magnetite-bearing samples. In contrast, unremagnetized limestones from Italy show hysteresis behavior which is more typical for magnetite-bearing rocks (Channell and McCabe, this issue). These results suggest that remagnetization in magnetite-bearing carbonates can be readily detected on the basis of hysteresis properties, prior to a full-scale paleomagnetic investigation.

Introduction

The paleomagnetism of Carboniferous limestones within the Craven Basin of northern England (Lancashire and Yorkshire) is treated in several published papers concerning Paleozoic apparent polar wander [*Turner*, 1975a,b; *Turner and Tarling*, 1975; *Turner et al.*, 1979; *Addison et al.*, 1985]. Although some of the sampled units showed anomalous magnetization directions, attributed to field reversals or early diagenetic processes, the predominant magnetization had SSW declinations and shallow up inclinations and was considered to be of depositional origin. However, more recently it has been shown that most of the Paleozoic carbonates in eastern and central North America were completely remagnetized in Hercynian-Alleghanian time (reviewed by *McCabe and Elmore* [1989]). These remagnetizations yield paleomagnetic poles near Late Carboniferous or Early Permian poles for North America and are exclusively of reversed polarity, consistent with remagnetization during the Kiaman Reversed Superchron. In the Appalachians, the remagnetization has been shown in several cases to be syn-Alleghanian [e.g., *McCabe et al.*, 1983]. Because previous studies of Craven Basin carbonates determined predominantly reversed polarity magnetizations and paleomagnetic poles close to Late Carboniferous-Early Permian poles for Europe, we suspected that these rocks may also have been remagnetized. This study was undertaken, in part, to test for remagnetization in these rocks.

We also attempted to expand the database of rock magnetic properties of remagnetized carbonates by including data for rocks in England and other areas outside of the Appalachians. Remagnetized magnetite-bearing carbonates in the Appalachians yield apparently contradictory results in standard rock magnetic tests for estimating the predominant domain state of the carrier magnetites. Some of these tests are not definitive indicators of domain state (e.g., Verway transition, Lowrie-Fuller test, maximum unblocking temperature). However, it has been suggested that the set of unusual test results could be diagnostic for remagnetization. The modified Lowrie-Fuller test [*Johnson et al.*, 1975] has indicated multidomain (MD) magnetite in several studies (e.g., Onondaga Limestone [*Kent*, 1979]). Other possible indications of predominant MD carriers in the Appalachian carbonates are maximum unblocking temperatures of about 500°C, considerably less than measured Curie temperatures (e.g., Helderberg Limestone [*Scotese et al.*, 1982]), and the common observation of coarse magnetite in magnetic separates (e.g., Helderberg Limestone [*McCabe et al.*, 1983]). On the other hand, single domain (SD) Lowrie-Fuller tests have been occasionally observed (e.g., Knox Dolomite [*Jackson*, 1990]), and the absence of the Verwey transition at about 120°K has been interpreted to indicate that very little of the magnetization is carried by true MD material in Helderberg samples [*Scotese et al.*, 1982; *Jackson*, 1990].

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Jackson [1990] pointed out that hysteresis results provide much more definitive constraints on domain state of the magnetite carriers than the tests discussed in the preceding paragraph, and he provided some very interesting new data. Hysteresis results and low-temperature demagnetization experiments [*Jackson*, 1990] indicated that the NRM in Onondaga, Helderberg, Trenton, and Knox samples resides mainly in SD material that appears to lack shape anisotropy. However, the "wasp-waisted" shape of hysteresis loops indicated that low coercivity material was present along with the dominant SD magnetite. This low-coercivity magnetite might be MD [*Jackson*, 1990], superparamagnetic [*Jackson et al.*, 1993], or a combination of the two. The conclusion that the Late Paleozoic remagnetization resides in SD grains has important consequences for choosing between chemical and thermoviscous remagnetization mechanisms [see *Kent*, 1985]. In addition, *Jackson* [1990] noted that the hysteresis ratios follow a trend rarely seen in synthetic or natural samples: both Hcr/Hc and Mrs/Ms are higher than expected, and together define a trend which is clearly different from those commonly found in other rock types and from synthetic samples containing mixtures of magnetite grain sizes [*Parry*, 1980, 1982]. The origin of these anomalous properties is not well understood [see *Jackson*, 1990; and *Jackson et al.*, 1993]. However, this suggestion of a unique rock magnetic fingerprint for remagnetized, magnetite-bearing carbonates bears further investigation. If these unusual hysteresis properties can be demonstrated in remagnetized carbonates from different regions and for different remagnetization ages, then hysteresis parameters might be used to determine whether remagnetization has occurred before a decision is made to undertake an expensive and labor intensive paleomagnetic investigation.

Geology and Sampling

Dinantian (lower Carboniferous) exposures in Britain represent a broad range of depositional environments and ecologic communities, including lime mudbanks, carbonate platforms, braided river sands, and deeper water limestone, chert, and shale deposits [*Leeder*, 1992]. The Craven Basin is one of several major fault-bounded Dinantian sedimentary basins in the

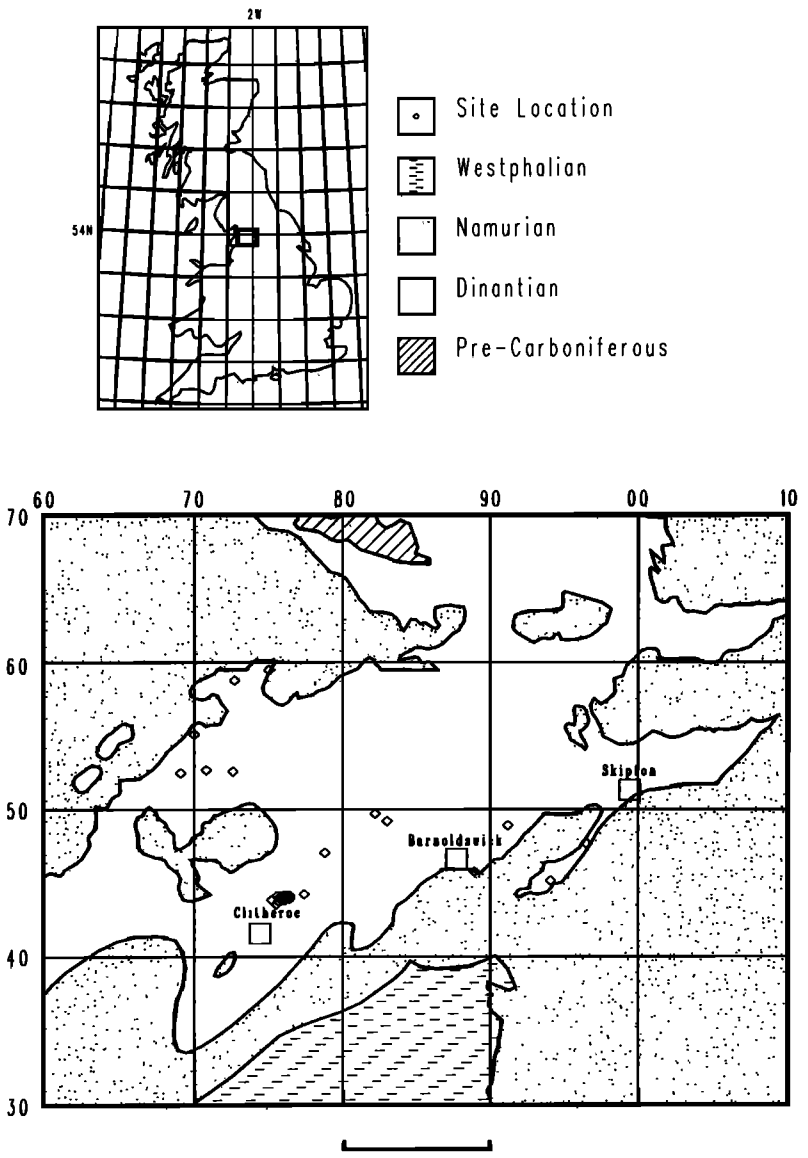


Figure 1. Generalized geological map of the Craven Basin showing site localities. The inset map of Great Britain shows the location of the study area. Scale bar is 10 km.

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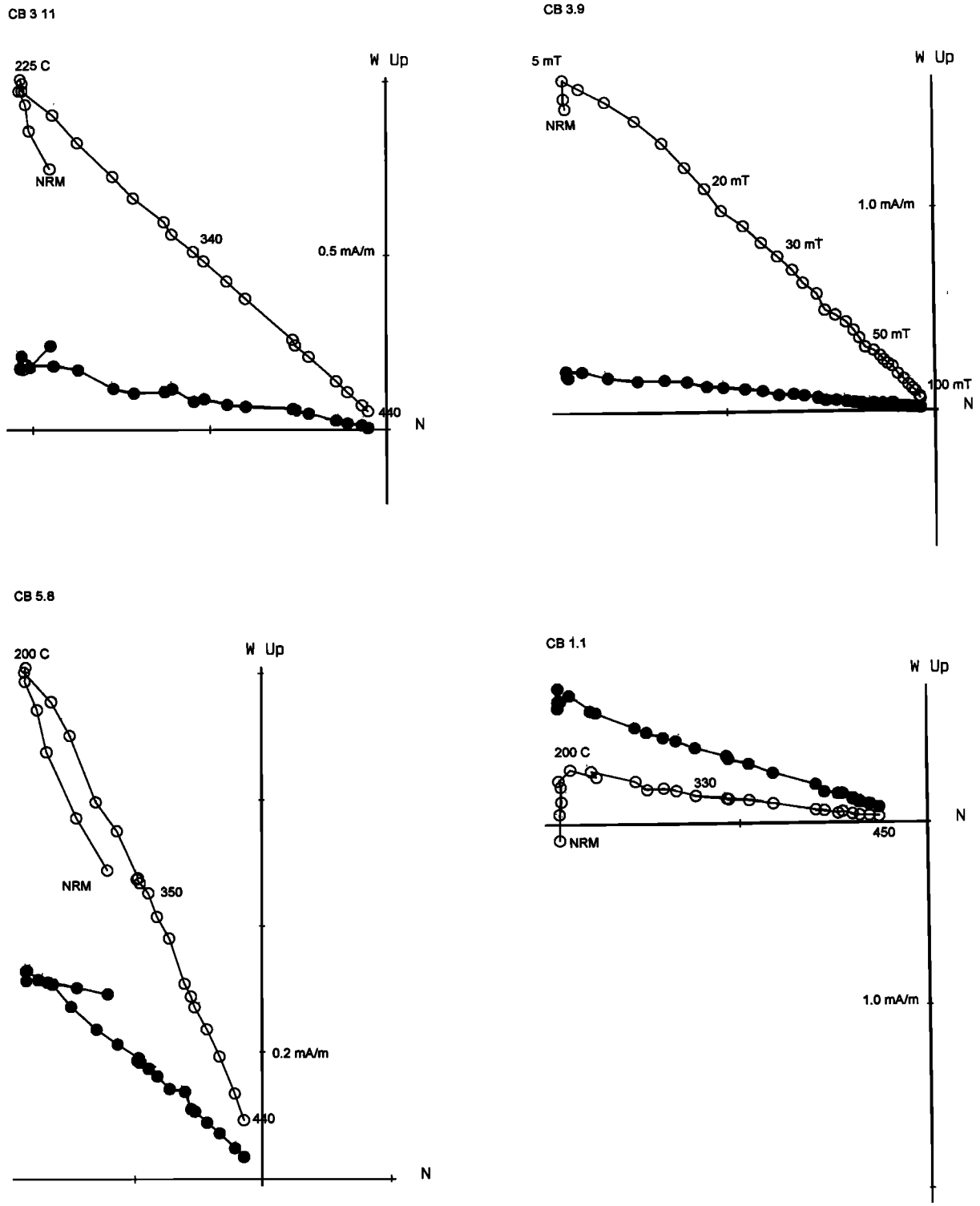


Figure 2. Demagnetization data for typical specimens from the Craven Basin. The top two plots show a comparison of thermal and AF demagnetization results for two samples from the same site. Data are not corrected for local tilt. Open circles are projections of the magnetization vectors onto the vertical plane; solid circles are projections onto the horizontal plane.

north of England. The sedimentary section includes marine carbonates and shales of Tournasian and Visean age. The succession, from oldest to youngest, comprises the Chatburn Limestone, the Worston Shale Group, and the Pendleside Limestone. The geology of the area has been described in detail by *Earp et al.* [1961]. A number of distinctive carbonate "reef knolls" may be found in the Clitheroe Limestone Complex of the Worston Shale Group, which show evidence for high (~30°) depositional dips [*Miller and Grayson*, 1972]. The Pendleside Limestone includes exposures of carbonate breccia-conglomerates, which have been interpreted as debris flow deposits associated with syn-sedimentary tectonic movements [*Gawthorpe*, 1987]. The

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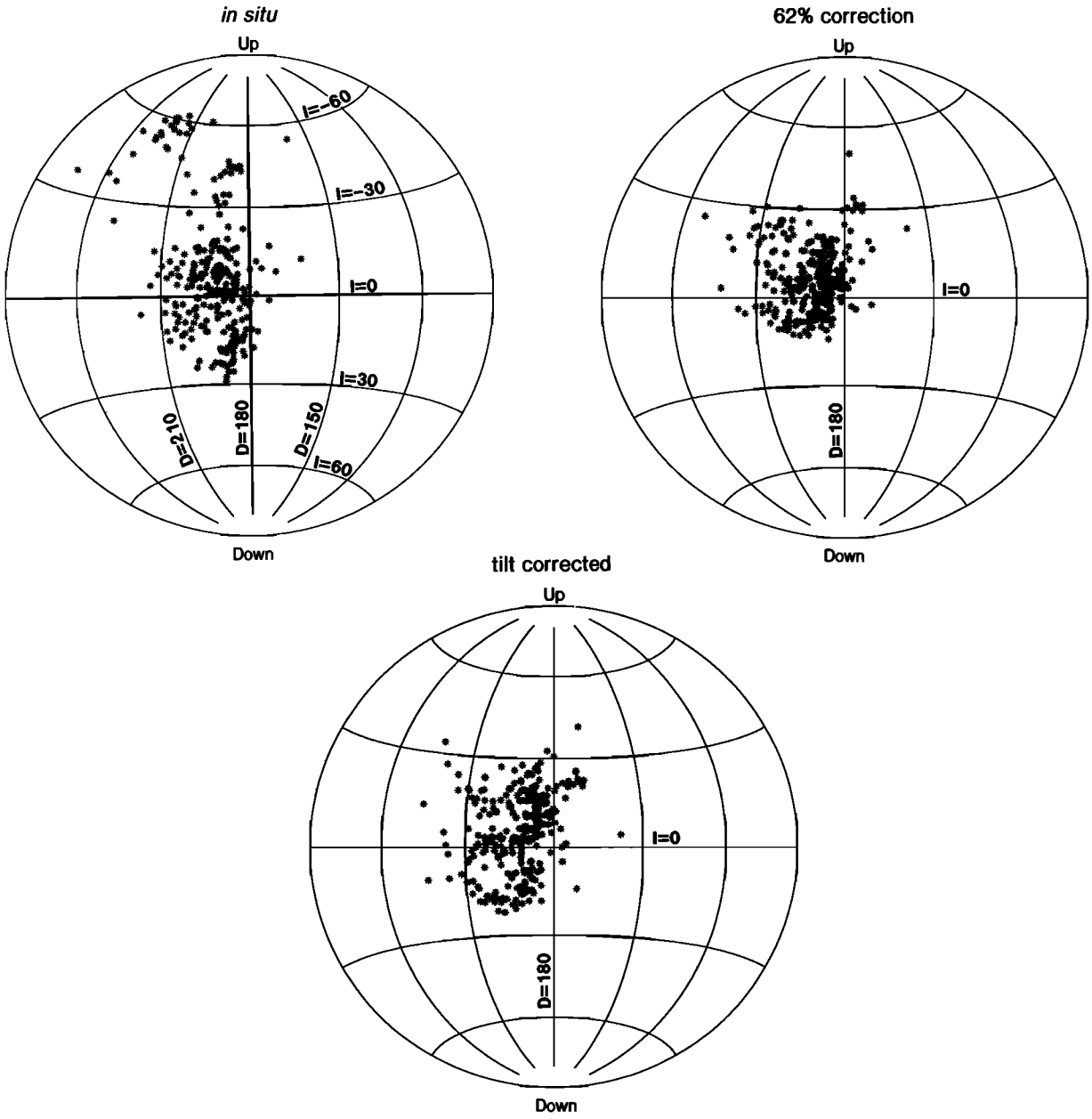


Figure 3. Specimen characteristic magnetization directions in equal area projection for the Craven Basin. Note equatorial projection. All directions plot on the upper (southern) hemisphere. The directions are plotted in three coordinate systems: before tilt correction (in situ), after a 62% tilt correction, and with the full tilt correction (see text).

sequence was deformed into a series of NE-trending folds in Hercynian (Late Carboniferous) time.

In order to investigate the possibility of remagnetization in the Craven Basin, we undertook a regional study of carbonate units within the basin. Sampling was confined to the Chatburn and Pendleside limestones and was designed for application of a regional fold test to bracket the magnetization age with respect to the age of folding. In addition, a synsedimentary conglomerate was sampled for a paleomagnetic conglomerate test. Although well-exposed in the area, we avoided the Clitheroe Limestone Complex: the reef flanks clearly have a depositional component of dip, and thus the paleohorizontal necessary for correction of paleomagnetic directions cannot be readily determined. Samples were taken from 20 sites in the Chatburn Limestone and eight sites in the Pendleside Limestone (Figure 1) using a portable core drill. One Pendleside site was in one of the distinctive synsedimentary debris flow breccia conglomerates.

Methods

Core samples were cut into standard 2.2-cm specimens for paleomagnetic analysis and taken to the paleomagnetic laboratories at either the University of Florida or Louisiana State University. Pilot alternating field (AF) and thermal demagnetization studies indicated that both methods resulted in excellent separation of the component magnetizations. All samples were subjected to stepwise demagnetization in 20 or more steps, and most were treated using the thermal method. At the University

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Table 1. Site Mean Paleomagnetic Directions of the Characteristic Magnetization

Site I.D.	Grid Ref (SO)	N/No	k	α95	In Situ D, deg	In Situ I, deg	Tilt Corrected D, deg	Tilt Corrected I, deg
			Chatburn Limestone					
CB 1	830 492	14/15	776.2	1.4	196.2	-6.7	198.0	18.0
CB 2	822 497	19/21	548.9	1.4	190.5	-7.4	190.9	14.7
CB 3	788 471	9/9	141.2	4.3	188.0	-45.7	172.4	-24.5
CB 4	752 439	14/14	167.5	3.1	186.6	17.4	186.0	-12.3
CB 5	758 441	9/10	645.9	2.0	218.2	-59.1	199.7	-1.2
CB 6	774 443	9/13	115.8	4.8	189.0	4.5	189.6	-21.2
CH 1	764 440	5/7	65.7	9.5	188.7	-2.6	190.8	-21.1
CH 2	764 441	7/7	207.9	4.2	188.5	-5.7	189.0	-15.4
CH 3	764 442	7/9	27.7	11.7	199.4	-43.7	185.0	-3.2
CH 4	762 442	8/8	467.2	2.6	223.6	-54.1	202.8	1.7
CH 5	760 442	7/8	24.2	12.5	231.4	-40.2	212.1	-1.8
CH 6	756 442	7/7	779.5	2.2	190.4	-32.2	184.5	-8.7
CH 7	761 439	6/10	79.6	7.6	181.8	13.2	181.4	-9.9
CH 8	760 439	8/8	373.7	2.9	187.0	11.7	187.8	-16.7
CH 9	755 436	6/12	95.0	6.9	181.1	-5.0	184.6	-25.5
CH 10	757 438	11/11	154.5	3.7	185.2	-1.1	186.5	-17.2
CH 11	912 489	12/12	33.5	7.6	197.9	12.0	197.3	-10.3
CH 12	941 452	8/10	185.7	4.1	199.1	8.2	201.4	-15.8
CH 13	691 525	8/8	37.3	9.2	191.1	-16.3	187.9	5.8
CH 14	708 527	8/8	61.9	7.1	191.8	5.2	191.6	-4.2
			Pendleside Limestone					
PL 1	889 458	9/11	87.7	5.5	203.9	3.5	208.4	-18.8
PL 2	965 477	15/15	162.7	3.0	196.7	-12.2	194.0	1.0
PL 4	726 526	7/9	50.8	8.6	198.7	5.3	198.5	-3.8
PL 5	700 551	11/11	257.2	2.9	204.5	-5.5	207.2	14.0
PL 6	750 595	10/18	160.4	3.8	184.0	-4.1	184.0	-14.1
PL 7	750 595	3/10	74.0	14.4	203.9	-11.4	204.9	-20.8
PL 8	727 588	17/18	285.5	2.1	188.3	-13.6	188.3	10.4

Means*	k	α95	D, deg	I, deg
In Situ	11.8	8.4	194.5	-10.3
full tilt correction	27.6	5.4	192.8	-7.6
62% correction	36.6	4.7	192.5	-8.9

Pole (62% tilt corrected data) latitude = 39.4°S; longitude = 346.0°E; dp = 2.4, and dm = 4.7

* $n = 27$

of Florida, samples were heated in a Schonstedt furnace, and remanent magnetizations measured with a three-axis 2G cryogenic magnetometer. At Louisiana State University, samples were heated in a three-zone furnace constructed from Lindberg components, and remanent magnetizations were measured with a three-axis superconducting magnetometer manufactured by CTF Systems. Demagnetization data were plotted on orthogonal demagnetization diagrams [*Zijderveld*, 1967], and component magnetization directions were calculated using the standard three dimensional least squares method [*Kirschvink*, 1980] after visual inspection of the diagrams. A magnetization was judged to have been sufficiently resolved if three or more data points followed a linear trend with a MAD value [*Kirschvink*, 1980] of 15° or less. Site mean and formation mean paleomagnetic directions were calculated using *Fisher* [1953] statistics. The fold test procedure of *McFadden* [1990] was used to determine the most appropriate coordinate system for analysis of the results.

Rock magnetic experiments were conducted at the Institute for Rock Magnetism in Minneapolis and at the University of Florida. Rock chips of about 50 to 100 mg were prepared from the core bottoms of Craven Basin samples for hysteresis measurements. Hysteresis loops were obtained on a MicroMag alternating gradient force magnetometer. For each hysteresis run, the coercivity (Hc), saturation magnetization (Ms), and saturation remanent magnetization (Mrs) were determined following correction of the moment data for the high-field paramagnetic or diamagnetic slope. After each hysteresis run, samples were given a saturation isothermal remanent magnetization (IRM), and demagnetized stepwise in DC fields. This allowed determination of the coercivity of remanence (Hcr) and a second estimate of Mrs.

The low-temperature behavior of Craven Basin samples was investigated to obtain further information on the properties of the magnetic carriers. A SQUID susceptometer (Quantum Design MPMS) was used to study the behavior of an IRM imparted at 10°K during warming through the pyrrhotite transition at 32°K, and the Verwey transition for magnetite at 120°K. The *Cisowski* [1981] test for grain interactions was applied on representative Craven Basin samples. This test involves the stepwise acquisition and AF demagnetization of IRM. Finally, the modified Lowrie-Fuller test for predominant domain state [*Johnson et al.*, 1975] was applied by comparing normalized AF demagnetization curves for IRM and anhysteretic remanent magnetization (ARM).

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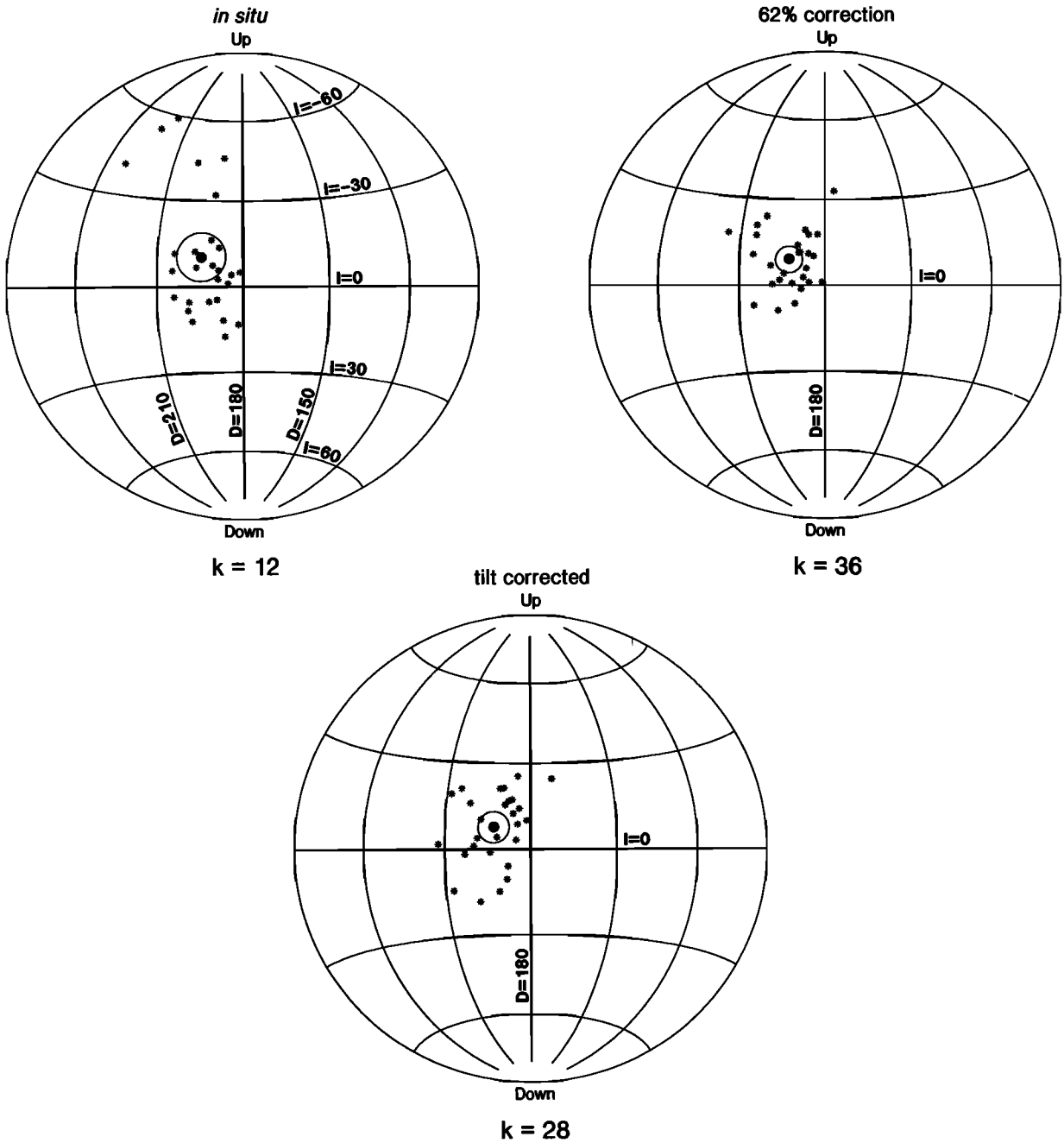


Figure 4. Site mean characteristic directions for Craven Basin (stars) plotted as in Figure 3. The mean directions (filled circles) are shown with their 95% circles of confidence. The best clustering (and minimum correlation statistic [*McFadden*, 1990]) occurs with a partial (62%) tilt correction (see text)

Hysteresis results were also obtained from some remagnetized carbonates collected in North America for comparison with the Craven Basin data. Hysteresis measurements were taken from samples of the Onondaga and Helderberg Limestones from New York State, two rock units featured in the hysteresis study by *Jackson* [1990]. Samples from the Mississippian Peratrovich Formation (Alexander Terrane, SE Alaska) were provided for this study by R.F. Butler. The magnetization in these rocks appears to have been acquired after a folding event of uncertain, but probably Mesozoic age (R.F. Butler, personal communication, 1993). Finally, J.W. Geissman provided samples of the Ordovician Poganip Group (Great Basin, Nevada) which carries a Late Paleozoic remagnetization [*Gillett and Geissman*, 1993]. The Alaska and Nevada samples are both from magnetite-bearing sedimentary carbonate units which are currently under paleomagnetic study.

Results

Paleomagnetism

The paleomagnetism of the Chatburn and Pendleside limestones is very similar to that of the Onondaga and other remagnetized limestones from eastern North America. Initial NRM

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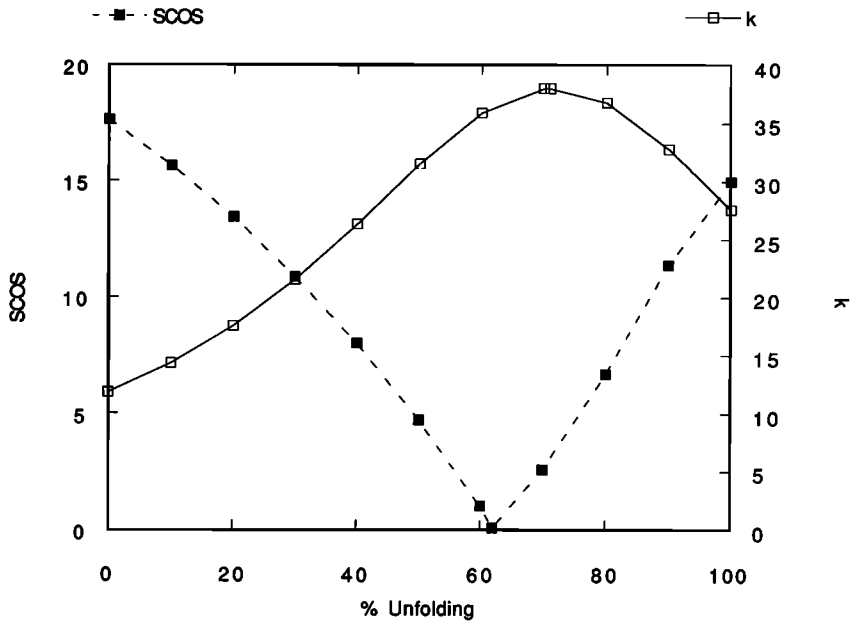


Figure 5. Incremental fold test for Craven Basin data. The correlation statistic (SCOS [*McFadden*, 1990]) reaches a minimum value at 62% unfolding, whereas the maximum k value is obtained after 71% unfolding.

magnetizations are rather strong, usually a few millamperes per meter, in contrast to the much weaker NRMs found in magnetite-bearing carbonates that have not been remagnetized [e.g., *Channell and Erba*, 1992]. Orthogonal demagnetization diagrams (Figure 2) show that a magnetization approximately aligned with the present field is removed below about 200° C. From about 200 to 450°C, a characteristic magnetization with SSW declinations and shallow inclinations (in stratigraphic coordinates) is removed, usually with linear decay towards the origin of the diagrams. The few samples that were subjected to AF demagnetization showed very similar trends, with most of the magnetization removed by AF treatment to 100 mT (Figure 2). The characteristic magnetization was well-resolved in most of the samples demagnetized, but some samples exhibited unstable behavior and did not yield usable data. Only one site (PL 3) showed unstable demagnetization behavior in all the samples demagnetized. Site PL 7, the synsedimentary conglomerate, yielded only three characteristic magnetization directions out of 10 specimens demagnetized. Although we consider these site data marginal at best, the fact that the SSW and shallow magnetization was found at all suggests that this magnetization occurred after conglomerate emplacement.

Figure 3 shows specimen characteristic directions in equatorial, equal area projection. Site mean directions are listed in Table 1 and shown in Figure 4. These site mean data were subjected to the fold test analysis method of *McFadden* [1990], including an incremental fold test (Figure 5). Figure 5 shows that the maximum concentration parameter k is reached at about 71% unfolding. However, the fold test based on k ratios [*McElhinny*, 1964] is not significant at the 95% confidence level for the maximum k direction compared with the fully unfolded direction. In contrast, the correlation statistic (SCOS [*McFadden*, 1990]) is significant at the 99% confidence limit for both the in situ and tilt corrected directions but is highly insignificant for the minimum SCOS statistic direction, which is reached at 62% unfolding. This result clearly shows the superiority of the *McFadden* [1990] fold test in determining the coordinate system which best approximates the structural setting of the strata at the time the magnetization was acquired. Figure 4 shows the site mean directions plotted in three coordinate systems: in situ, 62% tilt corrected, and fully tilt corrected. The best clustering of the directions occurs with a 62% tilt correction applied to the data. The results imply that the characteristic magnetization was acquired during Hercynian folding, well after deposition. The paleomagnetic pole calculated from the 62% corrected data (latitude = 39.4°S ; longitude = 346.0°E; dp =

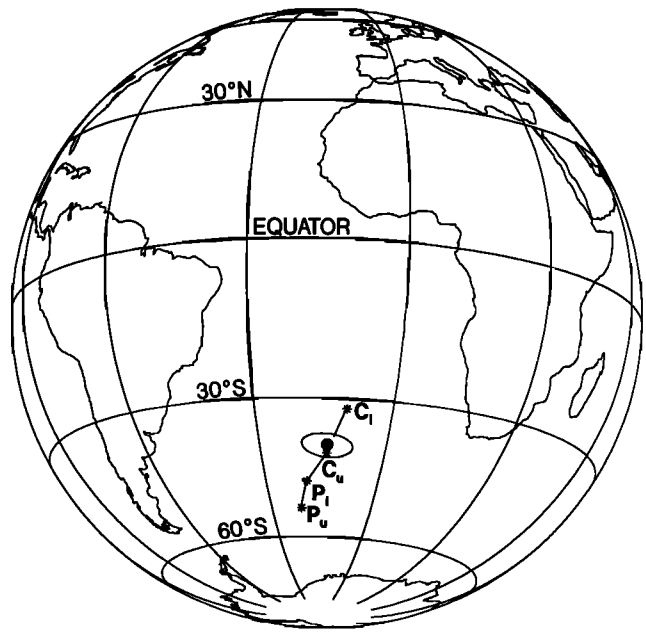


Figure 6. The Craven Basin mean paleomagnetic pole (62% tilt corrected data) is shown as a filled circle surrounded by its oval of 95% confidence (dp, dm). Paleomagnetic south poles for Europe (stars) for Permian and Carboniferous time are shown for reference [from *Van der Voo*, 1990].

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2.4°; dm = 4.7°) is plotted in Figure 6, along with Late Paleozoic poles for stable Europe. The Craven Basin pole plots very close to the Upper Carboniferous pole for Europe, fully consistent with the fold test result and exclusively reversed polarity of the directions.

Rock Magnetism

The behavior of a low-temperature IRM on warming to room temperature is shown in Figure 7 for a Chatburn Limestone sample. The steep decline in IRM from 10°K to about 50°K and its continued decline upon further warming suggests that superparamagnetic material makes an important contribution to the low-temperature IRM, consistent with the inferences of *Jackson et al.* [1993]. Note further that there is no indication of a pyrrhotite transition near 32°K, or of the Verwey transition near 120°K, similar to results obtained previously for most remagnetized units [e.g., *Scotese et al.*, 1982; *Jackson et al.*, 1993]. We infer from this that monoclinic pyrrhotite does not make a significant contribution to the magnetization in our samples.

Hysteresis loops (corrected for high-field paramagnetism or diamagnetism) from the Craven Basin limestones, the Onondaga Limestone, and the remagnetized carbonates from the Great Basin and from Alaska show remarkable similarities. Nearly all the hysteresis loops are wasp-waisted, indicating a bimodal distribution of magnetite coercivities (grain sizes). The examples shown in Figure 8 are typical and are similar to the examples given in *Jackson* [1990]. The most striking aspect of the data is in the hysteresis ratios. Figure 9 shows a log-log plot of the hysteresis ratios obtained from Craven Basin, Onondaga, Great Basin, and Alaskan samples. Remarkably, all plot along the same line but are quite different from expected trends for mixtures of magnetite grain sizes as shown. They are also quite distinct from the trends obtained from unremagnetized limestones from Italy [see *Channell and McCabe*, this issue].

Figure 10 shows the results of the *Cisowski* [1981] test, and for the modified Lowrie-Fuller test [*Johnson et al.*, 1975] for a typical Craven Basin sample. IRM acquisition (solid squares) and demagnetization (solid circles) cross at near 50% of the normalized moment, suggesting that the IRM is dominated by noninteracting fine particles, according to *Cisowski* [1981]. However, the test of *Johnson et al.* [1975] shows that ARM

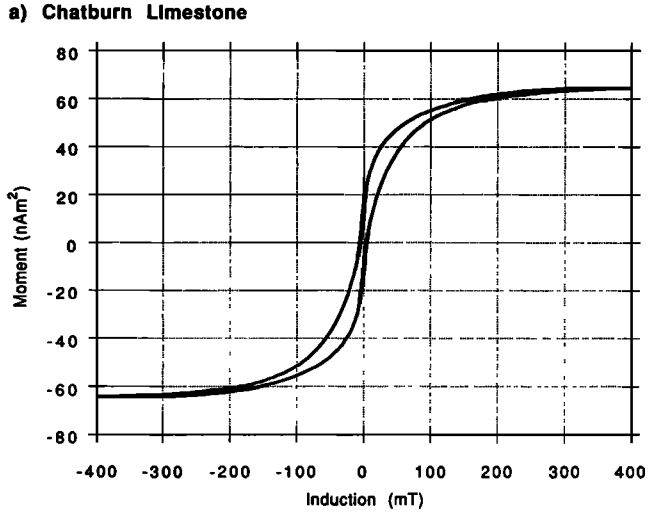


Figure 8. Typical hysteresis loops for (a) the Chatburn Limestone (Craven Basin), and (b) the Onondaga Limestone (New York). All hysteresis loops from these two remagnetized carbonates show "wasp-waisted" loops, consistent with the observations of *Jackson* [1990] for remagnetized Onondaga, Helderberg, Trenton, and Knox samples.

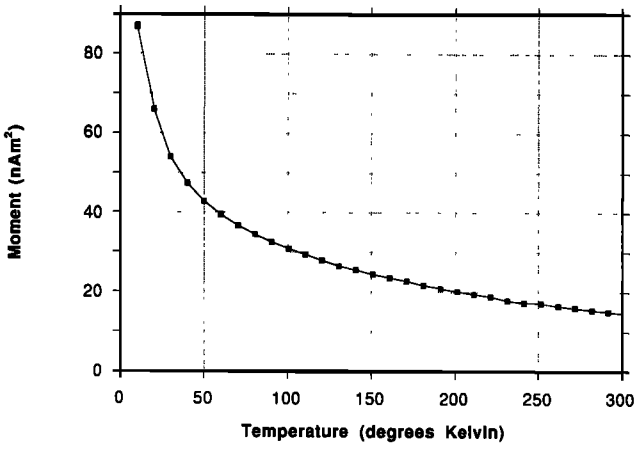


Figure 7. Decay of an IRM imparted at 10°K upon warming to room temperature for a sample of the Chatburn Limestone.

(open circles) is less stable than IRM during AF demagnetization, supposedly indicating the predominance of coarse (MD) magnetite. This is the same unusual result found in remagnetized Paleozoic carbonates from the Appalachians [*Jackson*, 1990].

Discussion and Conclusion

The phenomenon of Late Paleozoic remagnetization in the British Isles has been known for many years [e.g., *Chamaulan and Creer*, 1964; *Creer*, 1968]. Recent thermal demagnetization studies in a variety of rock types have indicated that the remagnetization event was quite pervasive, extending from near the Hercynian Front to the English Lake District and Scotland (reviewed by *Channell et al.* [1992]). Despite an early study in Ireland in which Late Paleozoic remagnetization was inferred for a Carboniferous limestone [*Morris*, 1971], these carbonates have not been generally regarded (in the published literature at least)

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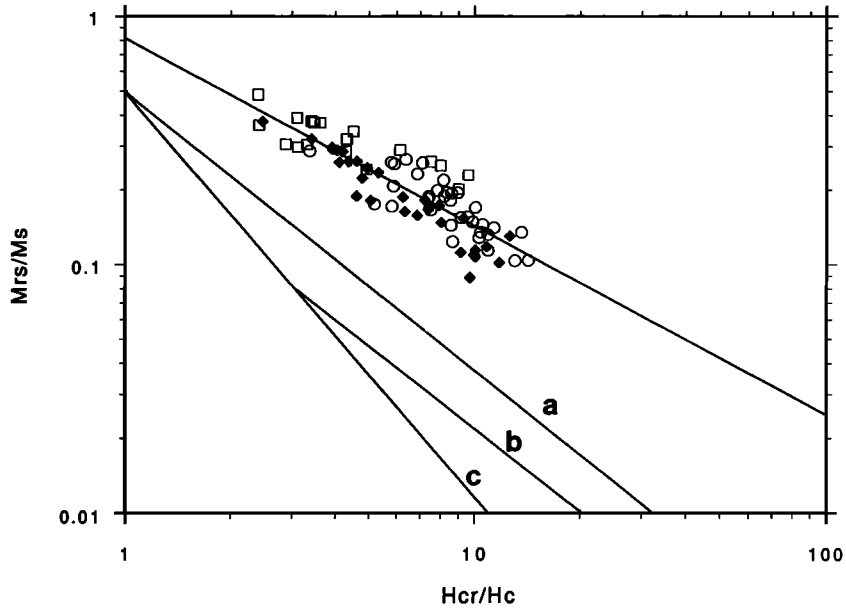


Figure 9. Hysteresis ratios from Craven Basin limestones (solid diamonds) compared with data from Alaska and Great Basin remagnetized carbonates (open squares) and the Onondaga Limestone (open circles). The best fit line to the data has a slope of -0.7546 and a y axis intercept of 0.82374, which is reasonably consistent with the results of *Jackson* [1990]. The results of *Parry* [1980, 1982] for synthetic samples are shown for reference: line a, SD + MD mixtures; line b, PSD + MD mixtures; and line c, single size magnetites.

as remagnetized [e.g., *Turner*, 1975b; *Addison et al.*, 1985]. However, the paleomagnetic data presented here for the Craven Basin clearly show that remagnetization of the "Appalachian" type has affected these rocks.

The rock magnetic data presented herein suggest that remagnetization in magnetite-bearing carbonates may be readily predicted on the basis of rock-magnetic data. Most diagnostic are the wasp-waisted shape of the hysteresis loops and the anomalously high Hcr/Hc and Mrs/Ms ratios. These results are consistent with those given by *Jackson* [1990] in his study of Appalachian remagnetized carbonates. The fact that remagnetized carbonates from the Appalachians, England, the Great Basin, and Alaska all share these unusual traits suggests that they may be diagnostic for remagnetization in this rock type, regardless of age of remagnetization or geographic location. This conclusion is strengthened by the observation that unremagnetized magnetite-bearing limestones from Italy have distinctly different, and more "typical", hysteresis properties [*Channell and McCabe*, this issue]. Previously, remagnetization could only be reliably documented with a full paleomagnetic investigation. We suggest that hysteresis data may now be sufficient to demonstrate remagnetization in magnetite-bearing sedimentary carbonates.

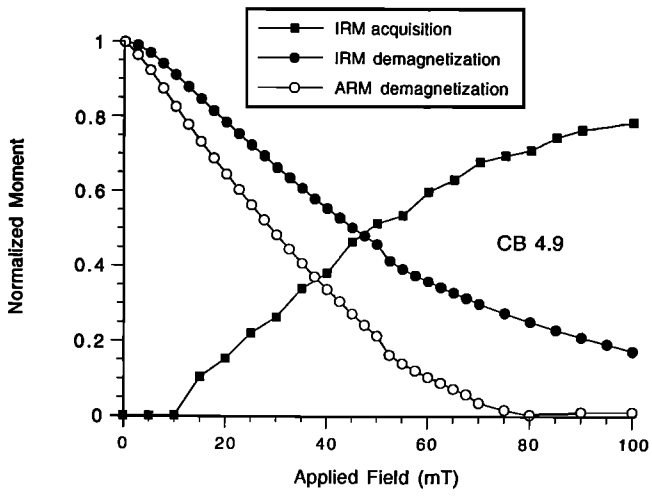


Figure 10. *Cisowski* [1981] and modified Lowrie-Fuller [*Johnson et al.*, 1975] tests on a Craven Basin sample. Acquisition (solid squares) and AF demagnetization (solid circles) of IRM; and AF demagnetization of ARM (open circles).

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